

The SpaceX Valuation: The “Bad Elden Ring Build” of IPOs - YouTube

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Transcript:

(00:02) Someone must extinguish thy flame. Back in 2022, Elden Ring released and became a smash hit success, selling over 30 million copies despite being a notoriously difficult and non-casual game. One of the 30 million who bought and played the game was Elon Musk, who was kind enough to post his character build on Twitter a few months after the game's release, which prompted everyone to fall out of their chairs laughing at its sheer stupidity.

(00:34) In short, his character makes no sense for this type of challenging game because it doesn't know what it wants to be, and it's not optimized for any particular outcome. He's wearing two shields and two swords, which is completely pointless since he can only use one at once. And since magic users shouldn't have shields anyway, this setup just slows him down for no real benefit.

(00:55) He's also a magic user, but he's wearing heavy armor that restricts his mobility, and it's not feasible to quickly switch armor in this type of game. Also, his health points are far too low for a character of this level, meaning that he'll still die quite easily despite the two pointless shields and the heavy armor.

(01:13) I bring up this anecdote because it's also a perfect way to describe SpaceX. It doesn't know what it wants to be and it's not optimized for any particular outcome. It's a company doing a somewhat random mix of things and losing money with most of them. The initial \$2 trillion asking price makes it the most overvalued company to ever go public, even more ridiculous than we work or group because at least they were aiming for much lower valuations.

(01:39) Most of the online discourse has focused on the company's historical results rather than its future prospects. But all corporate valuation is based on a company's future cash flows and how its market size, market share, and unit economics translate into those cash flows. So, I've spent around 50 hours reviewing the SpaceX S1 filings from around 100 comparable companies across other industries and plenty of market data, all to create a detailed model that illustrates why this valuation is absurd. The bottom line is that even if

(02:10) you assume massive outperformance for over 20 years, the math still doesn't math. Now, having said all that, I would not recommend shorting the stock because we're in an extremely dumb market and overall timeline in which fundamentals no longer matter, meaning SpaceX will probably hit \$5 trillion right after going public.

(02:30) You can get the full valuation model for free by following the pinned link below this video. You can also find a written summary there. Or if you want to keep watching, I will take you through the overall methodology, the comparable companies, and the forecast for each part of the SpaceX business. Okay.

(02:46) So, I'm going to start with how you value SpaceX and the overall methodology we're using. Then, we'll go through Starlink and the forecast and the valuations for that division. Then, we'll look at the AI or XI division and look at the forecast and valuation there. Then, we'll go through the space division, the forecast and valuation there.

(03:04) I know that you will probably have some objections after I cover all this. So, we will discuss some other possibilities like orbital data centers, lunar tourism, trips to Jupiter, Mars, the list goes on. And I'll explain why I don't think these really make a material impact on the company's current valuation.

(03:21) And then I'll conclude with some final thoughts and why I am staying away from this offering. So, let's start with how you value SpaceX at a high level. As you probably know, the company operates three different divisions with very different growth, margin, and capex profiles. So, you have Starlink, which is essentially a high- growth telecom company that is focused on emerging markets and also has a good amount of enterprise revenue from ships and aircraft that are using the service.

(03:47) Then you have AI or XAI. And then you have the space division. I couldn't find a separate logo for it, so I'm just listing the SpaceX logo right there. Since each one is very different financially, we think the best approach is to use a long-term sum of the parts DCF where we apply a separate DCF to each division.

(04:07) We use separate discount rate and separate terminal value assumptions and then we sum up everything at the end and make some adjustments to figure out what the overall company is worth. Now, we could potentially use valuation multiples for at least some of these divisions. For example, we could apply high growth telecom multiples to Starlink, but I haven't done that here because the high growth potential and the very wide range of potential outcomes here means that valuation multiples are probably not the best approach.

(04:34) For each one of these divisions, I have assumed an initial state, which is roughly where it's at right now, and then made it progress to a more mature state over the span of 20 years because that's the forecast period in each DCF. So as a few examples of these progressions for Starlink, I have looked at high growth satellite and emerging markets focused telecom companies here.

(04:57) I think that Airtel in India and then the MTN group in Africa are probably the two best comparables for where Starlink is at right now because they're both high growth in

the double digits in terms of subscription growth, revenue growth, things like that. And similar to Starlink, they're targeting markets with a lot of geographic overlap.

(05:17) Now, after 20 years, Starlink will presumably look something more like T-Mobile or Vodafone or one of these other mature telecom companies. Today, if you look at the XAI division or AI division, I looked at companies like Coreweee and Nebius because effectively the AI division is doing something similar, renting out its unused compute capacity.

(05:36) I also looked at some data center providers. Unfortunately, here there aren't really any mature public companies in this space, so I had to use a bit more guesswork here than I would have liked. with the space division. I looked at companies like Rocket Lab and MDA and Redwire and Firefly, these high- growth rocket and space technology companies and then assumed that after 20 years the profile starts to resemble more the big five defense contractors in the US companies like Boeing and General Dynamics and Loheed Martin for example.

(06:07) So that's the overall approach. Let's go through Starlink now and I will pull up the Excel part for Starlink here and go through some of that. But I want to start with my highlevel impressions of this division first. So Starlink is important because it accounts for most of SpaceX's revenue, operating income, and unlevered free cash flow.

(06:27) And unlike the other divisions, it's actually profitable. So it's quite important here. Now I think the closest analog as I was saying before are probably these higher growth larger emerging markets focused telecom companies such as Barti Airtel in India or the MTN group in Africa. In general with Starlink the subscriber growth is very high.

(06:50) The market penetration keeps going up but the average revenue per user keeps falling because the company keeps targeting more and more emerging markets customers. And to increase their market penetration, they're also offering pretty generous discounts on Starling packages. And you can see that pretty well if you go and look at the historical data here where the subscriber growth is pretty amazing going from around 2 or 3 million to close to 10 million just in the span of 2 years.

(07:16) However, their monthly average revenue per user is also falling. It fell from about 99 to 81 in that same time frame. One major disagreement I have is that the S1 cites a \$660 billion addressable market based on 1.8 billion households. But this is obviously ridiculous because there aren't that many rural households that do not currently have internet and are also in a position to pay for Starlink internet.

(07:40) So I have adjusted this down quite a bit and instead of 1.8 billion, I am going with the assumption that the addressable rural households are more in the 400 million range. And frankly, I think even that might be a little bit too aggressive, but I think it is probably closer to the truth than the 1.

(07:56) 8 billion number that they provide. So, looking at this at a high level, in the optimistic case scenario, right here, we have the total subscribers going to close to 200 million by the end of this 20-year period. The average revenue per user falls over this time, although it still remains pretty healthy and well above what many of these companies are at right now.

(08:17) In the enterprise segment, they go from approximately 30 to 40,000 ship customers currently to more like 150,000 by the end, which is about 60% of the addressable market here. And the overall revenue goes from around 11 or 12 billion as of the end of the most recent fiscal year to more like 76 billion by the end of this 20-year period.

(08:44) So, there's obviously some pretty astounding growth there. Now, if we go down and take a look at the margins and the cash flows with Starlink, one of the issues is that they're already operating at a pretty healthy IBA margin of around 60%. So, there is probably some room to grow this and I do assume some expansion over time to more like the 74% range, but the capex remains relatively high over this whole period ending at about 16% of revenue.

(09:09) And so, the unlevered free cash flow margin here ends up at around 41 to 42%. And it does grow by a lot, but maybe not by quite as much as you expect because it is a telecom company. It is launching satellites and all that is quite expensive. If you go and compare this to some of the comparable companies such as the ones in India and Africa that I was just referring to, you can see that they have subscriber bases that are in the 200 million, 300 million up to 6 or 700 million range.

(09:36) And then if you look at their capex figures, they're in a pretty similar range. 13 14% of revenue, 21% of revenue. They also have double-digit growth rates and their ebdom margins are also quite healthy around 40 to 50 to 60% right here. So that's where these starlink assumptions are coming from. Now once I had those in place, I went down and also created a few sensitivity tables in the main tab on different factors such as the market penetration and the number of addressable rural households in the beginning and other things like that and got to a rough

(10:06) range here of about a 100 to 400 billion for just Starlink. I have this table here that summarizes the output. The main difference in all these cases is the number of individual subscribers. The number of ship and aircraft customers is also quite different ranging from 100,000 to 200,000 over this time frame depending on how the company performs.

(10:27) And this creates revenue differences where the final revenue could be anywhere from 51 billion to 102 billion. The EBID could be anywhere from 38 billion to 76 billion. And overall that translates into valuation differences here where on the low end, what I call the realistic case, we're looking at maybe an enterprise value of about 150 billion.

(10:47) On the high end, it might be closer to 300 to 350 billion. And that's what Starlink looks like overall. And I've listed some of the medians from the current comparable companies over here on the right side for your reference. Let's go to the AI division now and talk about the forecast and the valuation here.

(11:04) So this one's a little bit confusing because there are actually three separate business lines. You have Twitter or X which earns money from ad revenue and also from data licensing. Then you have the AI solutions and infrastructure division which sells AI subscriptions to Grock and also sells tokens and charges a per token rate similar to what Open AI and Anthropic are doing for their enterprise customers.

(11:27) And then they also rent unused compute capacity which corresponds to the recently announced Anthropic deal. So you have a couple different things going on here. And so the overall approach is that right now the company has about 1 gawatt of data center capacity. So we've split this up into a portion that it's using for its own purposes for training and inference and then a part that it's renting out to anthropic and earning about 1.

(11:50) 25 billion per month from and presumably in the future if that is successful it will grow and expand other companies. And for each of these we've assumed dollar per megawatt rates. Now with the overall AI data data center market, it was about 31 gawatts globally as of the end of fiscal 2025. And we assume about a 10-fold increase over the next 20 years.

(12:12) So that it goes to just above 300 gawatt by the end of fiscal 2045. And we assume that XAI goes from around a 1 or 2% share right now to more like 4 to 6% over this time frame. With the other assumptions, we have some very moderate growth in the dollar per megawatt figures and also the number of X or Twitter users and the ad revenue there.

(12:33) So, let's pull up the tab for the AI segment and I'll take you through some of the numbers here. So, you can see that the topline driver is simply the size of the AI data center power capacity market right here, 31 gawatt, and this increases to more like 312 by the end. And then there's the name plate compute draw which is essentially the amount of capacity that XI has and this goes up to about 15.

(12:56) 6 in our optimistic case right here. We assume that overall they use about 80% of this for themselves and then they rent out about 20% although it starts out at a higher number because we're assuming here that they're renting out about 500 megawatts of capacity to Anthropic initially. And this then explains the 10 billion in revenue that we project they'll earn from Anthropic in 2026.

(13:18) This then increases to more like 20 to 30 billion after this as they get a full year of the Anthropic deal and presumably expand other companies. Overall revenue-wise

here, they go from about 3 billion to closer to 221 billion by the end of 20 years, which is obviously in a very aggressive forecast. But frankly speaking, that's really the only way the numbers here work.

(13:41) Now, right now, this AI division is losing a massive amount of money, negative 183% for the operating margin. But we do assume that by fiscal 2030, they will become operating income positive, which may or may not happen. I picked this date because that's also the date on which Open AI and Anthropic appear to be projecting some profitability in their businesses.

(14:03) So we do something similar here and we assume they scale up quite aggressively. The overall EBID margin here goes up to about 69 or 70% by the end but capex remains relatively elevated and so the unlevered free cash flow margin is around 40% which is actually pretty close to what it was in the Starlink division as well. Now I know that some of these margin and growth rate assumptions may seem aggressive.

(14:24) However, what I would point out is that if you look at NeoCloud companies like Coreweave and Nebius right now, they actually do have pretty high EBID margins of 41 to 58% and even many of these smaller companies do as well. The problem is that they're all spending an absurd amount of money on capex. So, their cash flows are very negative, at least right now.

(14:43) But my point is that these numbers are grounded in something. And the same goes for the capex per new megawatt of capacity for many of these companies. We're pulling all these numbers directly from their filing and even trying to make some estimates for the average revenue per megawatt of capacity here as well.

(14:58) So that's the AI division. Now summing up everything here, we can see that the active capacity differs pretty strongly in the different cases. So does the unused capacity and then that creates pretty big differences in revenue, although maybe not quite as big as you might expect. 180 billion to 263 billion is actually smaller than the Starlink differences percentage-wise.

(15:21) The EBA is also quite different. And with the valuation ranges here, the main reason these ranges are so wide is because the discount rate is quite wide. It starts at around 15% and falls to 10% based on the comparable companies. And so we get a very wide range of values here. But I would say at the low end, you're probably looking at a 50 to 100 billion valuation for this division.

(15:45) In the middle of the range, maybe a \$150 billion valuation. And at the top end of the range, maybe closer to 200 to 250 billion, something like that perhaps. And again, I've linked some of the stats from the median comparable companies over here on the side. So, let's now go to the space division. Now, this one's ironic because SpaceX, as the name implies, is best known for its rockets.

(16:07) The S1 filing has tons of pictures of rockets taking off and different planets and the solar system and all that, but it's actually the least important division valuation wise because it's growing much more slowly than Starlink or XAI. And the cash flows and profits are pretty marginal despite this much lower growth.

(16:26) One point of confusion here is that the S1 discloses the data and KPIs in a bit of a confusing way. And what I mean is that if you go in and look at this, they disclose these figures about mass to orbit and the number of launches. But what they don't tell you is that most of these launches are actually for Starlink satellites.

(16:45) So they're not to external customers. Therefore, in our version of this model, I went in and actually manually adjusted this and brought down the number of orbital launches because I just want to focus on external customers because their space revenue here doesn't include any revenue that Starlink is paying to launch its satellites to this other division.

(17:04) This is all revenue from external customers. And so I've adjusted down the orbital launches to reflect this. Overall, we've assumed some pretty rapid expansion in the number of orbital launches from one launch every 10 days to just over one launch per day by the end of a 20-year period. In our base case, what I call the optimistic case here.

(17:24) We've also assumed some pretty significant margin increases and capex reductions over time. So, let's go through these numbers. We're starting off at around 40 to 50 orbital launches per year for external customers. This goes up to more like 400 by the end of the period. Total space revenue right now is at about 4 billion in the most recent fiscal year.

(17:44) But because the number of launches go up by so much and because they keep earning more and more from each launch. They're earning about 100 million right now, but by the end of 20 years, I'm assuming this goes up to more like 300 to 400 million, we get to about 150 billion in revenue by the end of 20 years. Now, the cost of revenue, the gross margins here don't really change that much over time.

(18:04) They decrease very slightly because they mentioned something about that in the filing. The main difference really is that the operating expenses fall quite significantly. So does the capex. And so right now the company is sort of just barely negative in terms of operating income. Actually more than just barely negative.

(18:21) It's a negative 15% margin, but the two years before that they were kind of borderline in terms of operating income. We assume that over time they get up to around a 40% margin here. Capex does remain somewhat significant, but it sort of falls to the level that a lot of the big five US defense contractors are in and goes down to about 5% of revenue here.

(18:42) That said, the overall operating margin is simply lower than it is in the Starling or AI divisions here by the end. And so the unlevered free cash flow margin is also lower. It's at around 28% by the end of this period versus more like 40% in the other divisions. The EBITD margin is also lower more like 40% versus the 60 70 or 75% that we saw in the other divisions.

(19:06) If you look at the overall valuation output here, you can see that the main differentiator in these different cases is the number of external orbital launches. Revenue differs by quite a bit as a result of that and EBITD also differs by quite a bit. 41 billion versus 97 billion across the cases. In terms of overall valuation here in the low end, we're talking about perhaps around 50 60 70 billion, something like that.

(19:31) The middle of the range, maybe around 130 to 150 billion, something in that range. And then if you go to the top end of the range, it's perhaps more like a \$200 billion valuation for this one division. So let's step back now and do a quick summary of this valuation and what we've found so far. The bottom line is that even in this optimistic case, it's really hard to see how the numbers here add up to anything close to two trillion.

(19:55) So, I have a graph here with the sum of the parts DCF valuation and the output in this case. And you'll see that we get up to around 400 to 500 billion here, which is great except the asking price is somewhere between 1.5 and 2 trillion as of the time I'm recording this video. And you can go into my Excel file here and look at some of these graphs and try changing around and you'll see that it's just really really difficult to get anywhere close to the initial asking price in this case.

(20:20) Now, if you disagree with some of my assumptions, that's perfectly fine. You can go into the Excel file and change them around yourself. But what you'll find is that even if you tweak things like the orbital launch growth rate or the AI data center market share or the market penetration for Starlink or anything like that, yes, the numbers do change, but for the company overall, we're not talking about massive changes here.

(20:44) It's still in this rough range of about 300 to 400 to 500 billion, something like that. And so based on all that, I find it really, really hard to buy into anything close to this proposed valuation. Now, I want to address some objections here because I know that if you're a fan of Elon Musk or SpaceX or Tesla or all those other various ventures, you are probably saying right now, well, wait a minute.

(21:06) You're looking at the company as it is today, but what about those orbital data centers or lunar tourism or trips to Mars or faster than light travel or asteroid mining? And you could keep going. Let's talk about contact with aliens, with extraterrestrials. Maybe we can add Jupiter tourism to this list as well. And indeed, the S1 actually mentions some of these future markets, point-to-point terrestrial travel, space tourism,

inorbit manufacturing, passenger and cargo transport to the moon and Mars. Now, all these are very

(21:34) cool and interesting ideas. But the problem is that they're so speculative and so expensive and so far away that they're impossible to factor into a valuation today. Even if you assume that some or all these happen by the year 2050 or 60 or something like that, from a present value perspective, factoring in the discount rate and the time value of money, they're probably not worth all that much today.

(21:57) And we have so little detail about these that they're not possible to include in a reasonable valuation. I think the more relevant point here is the fact that NASDAQ just changed its rules and made it much easier for various indices to include SpaceX, which a lot of people object to. Now, I'm not going to say that they're trying to dump the stock onto a bunch of retail bag holders.

(22:18) I will label this retail investors instead. But I will say that it really does raise questions how they suddenly change the rules about what it takes for a company to be included in an index all based on the fact that we have this one big company and several other big companies later this year coming up with these potential IPOs. I'm going to wrap up here and give you some of my final thoughts about the company and why I'm staying away from this particular offering.

(22:43) I realize this video has come across as perhaps a little bit critical of Elon Musk, but it's really not meant to be a personal attack on him. Now, I still think his Elden Ring build is hilariously bad and I would never do anything like that. But the reality is that I think SpaceX is actually a pretty inspiring company and Starlink is exceptional.

(23:00) It's really impressive what they've done. They've built one of the biggest telecoms around in really only the span of a few years, although it's longer than that if you count the founding date of SpaceX as an entire company. I'm less impressed with their AI division and it remains very much to be seen what the economics there look like.

(23:19) The question here though is price versus value. And going back to what I said in the beginning, the math doesn't math. I think what it really comes down to is how you want to think about your investments and how you want to play them. Do you want to go after them in a random and disorganized way without much of a coherent strategy or do you want to optimize your investments in your portfolio for specific outcomes? I would put SpaceX in the category of a side quest.

(23:45) It's something that you might put some money in to play around with if you're bored and you want to do it, but I certainly wouldn't rely on this company for anything substantial. I wouldn't put all your hopes in this company. And if you do want to invest, that's great, but just be aware of what you're getting into and what it's actually worth, even with some very, very optimistic assumptions.

(24:05) That's about it for this video. So, let's do a recap and summary now. How do you value SpaceX? We think the best approach is a long-term sum of the parts DCF where you value each division separately based on these 20year forecasts and then add up everything at the end. We went through Starlink forecast and valuation.

(24:22) Overall, they are greatly expanding their market penetration. They're getting a lot more subscribers and enterprise customers, but their rates at least for the individual customers are also falling over time even as the enterprise rates are perhaps staying about the same or rising. Overall, we think that a valuation somewhere between about 200 and 300 billion might be reasonable for Starlink.

(24:43) For the AI division, it's very speculative, but they appear to be doing a combination of selling their own AI tools and services, renting out some of their unused compute, and then earning advertising revenue from X or Twitter. There's a very wide range of outcomes here, but a reasonable case might be that it's worth about a hundred billion dollars, but honestly, this could be anywhere from zero up to a lot more than that.

(25:07) With the space forecast and valuation here, it's tricky because you have to separate out what they're doing for Starlink versus what they're selling to external customers. The bottom line though is that it's probably worth roughly the same as the AI division, but with a lot less variance just because it's been around longer.

(25:22) the economics are a little bit better known, the growth rates are lower, we have a better idea of what to expect there. If you doubt all this and you think that the company is actually worth a lot more than four or 500 billion or something in that range, and you point to things like orbital data centers or lunar tourism, I would say that's great and maybe it will be a factor in the future, but it's going to be a factor so far into the future that it's not really relevant for the valuation today.

(25:45) Overall, I don't necessarily think that this is a bad company, per se. Like I said, I think it's very inspiring. I think Starlink is very impressive. But for me, it just comes down to price versus value. And I think this is just so overvalued right now that I wouldn't buy into this as it stands right now. But your mileage may vary.

(26:05) And I think it's just important to be informed as a company like this goes public. So that's it for our valuation tutorial. Hopefully now you know a little bit more about how to think about a company like this and what to do with valuation when a large company like this operates some very very different divisions and what that means in terms of the overall company's value.